

Retail Sales Data Cleaning and Exploratory Analysis

Bluestock Fintech Internship - Week 1 Core Data Analytics Foundation

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Purpose

This report documents the cleaning, validation and exploratory analysis of a simulated retail sales dataset. It summarises data quality, sales performance, customer and product patterns, statistical measures and practical business observations.

Data disclosure: The analysis uses synthetic training data and contains no Bluestock customer, financial or operational information. The findings demonstrate analytical technique and must not be interpreted as Bluestock business performance.

Completed orders	Total revenue	Total profit	Profit margin	Average order value
323	INR 1,164,271	INR 397,041	34.1%	INR 3,605

Executive summary

- The cleaned dataset contains 500 unique orders. 323 are completed and included in financial KPIs.
- Completed orders generated INR 1,164,271 in revenue and INR 397,041 in profit, producing a 34.1% margin.
- Electronics is the highest-revenue category, while North is the leading region.
- Travel Backpack is the highest-revenue product at INR 165,475.
- The peak month is 2025-10 with INR 123,896 in revenue.

Analysis framework

Input	Cleaning	Analysis	Outputs
506 raw records	500 valid records	KPIs, trends, statistics	Clean CSV and EDA report

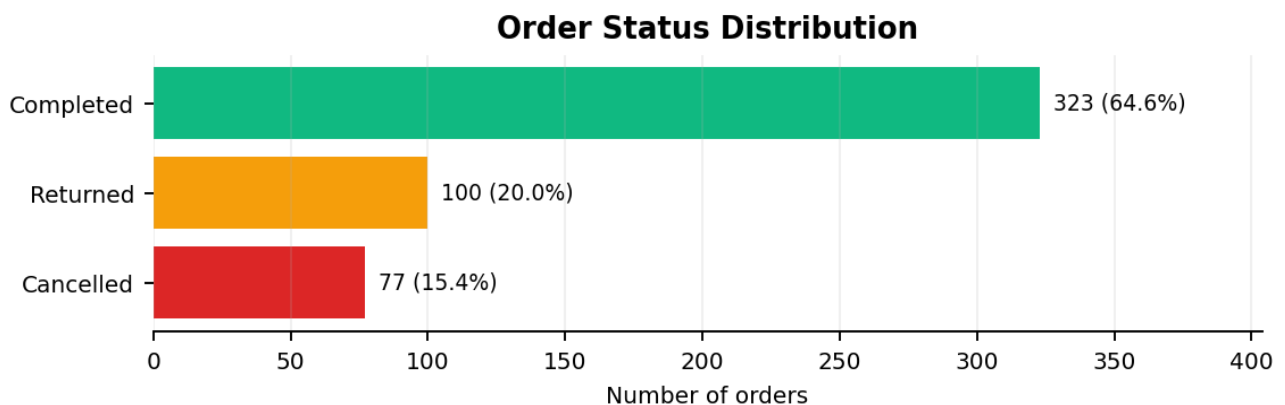
1. Data Cleaning and Validation

The raw file was profiled before any transformation. Each identified issue received a specific correction, and the same checks were repeated after cleaning.

Quality check	Raw	Clean	Treatment
Records	506	500	Removed six exact duplicate rows
Exact duplicate rows	6	0	Dropped repeated records
Missing customer names	6	0	Recovered names from Customer ID
Missing discounts	4	0	Replaced blank values with 0%
Non-standard category labels	1	0	Mapped Product ID to the canonical category
Region labels with extra spaces	3	0	Mapped each city to its canonical region
Non-standard UPI labels	2	0	Standardised the payment label to UPI

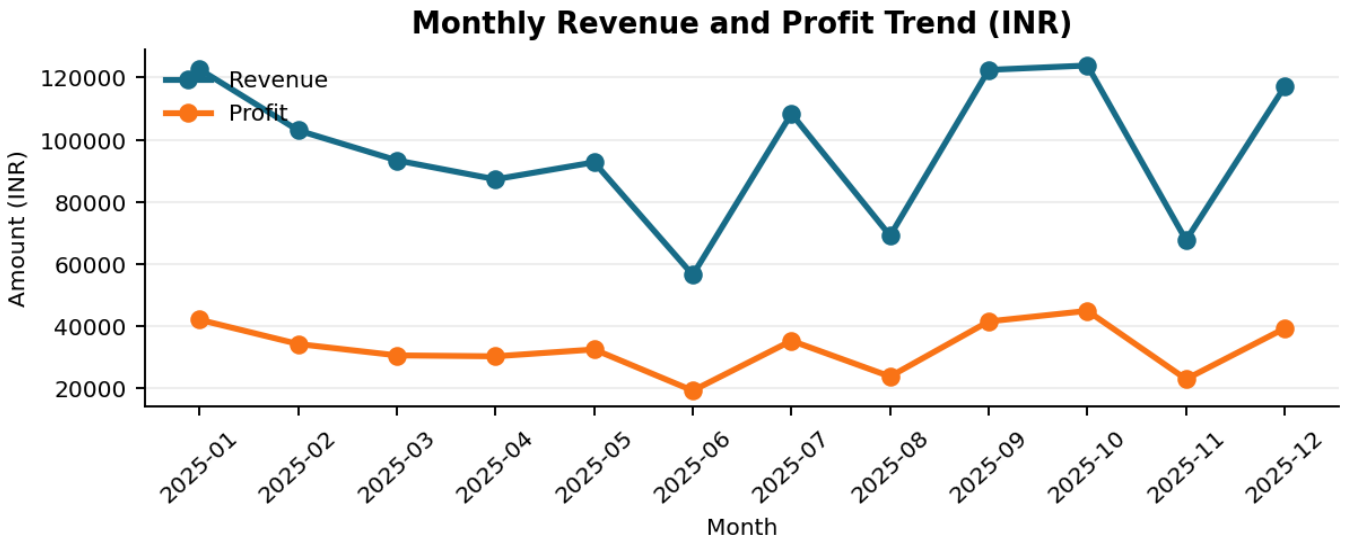
Financial treatment

Revenue, total cost and profit are set to zero for Returned and Cancelled orders. This prevents unsuccessful transactions from overstating performance. Completed orders are used as the denominator for average order value.

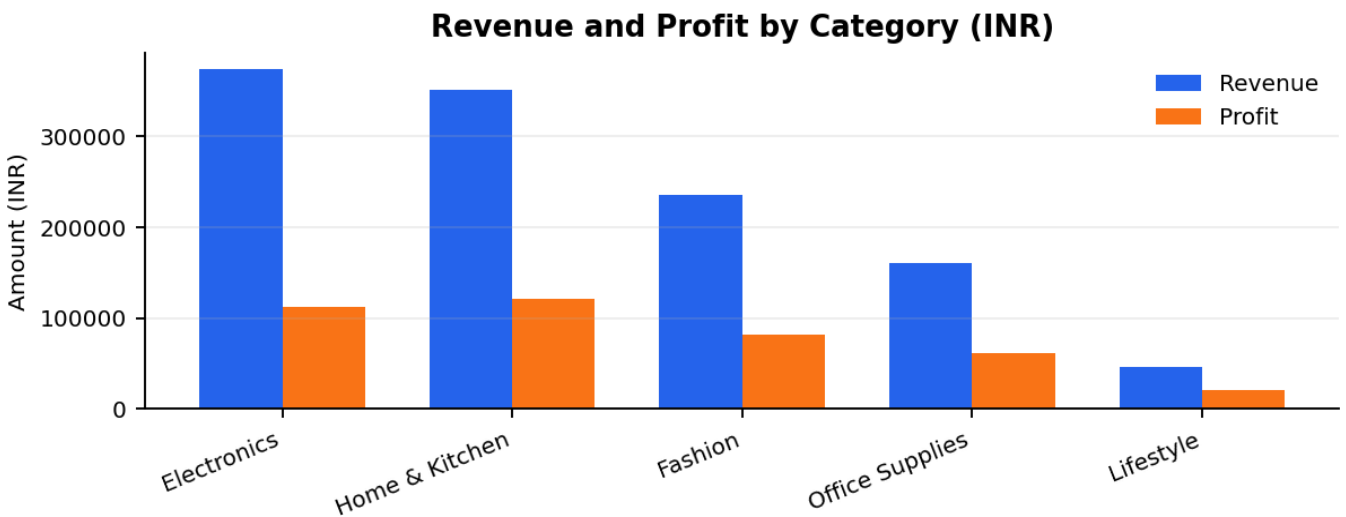


Validation result: The cleaned file has zero missing values, zero duplicate Order IDs, valid discount ranges and consistent customer, product, category, region and payment labels.

2. Sales Performance and Trends



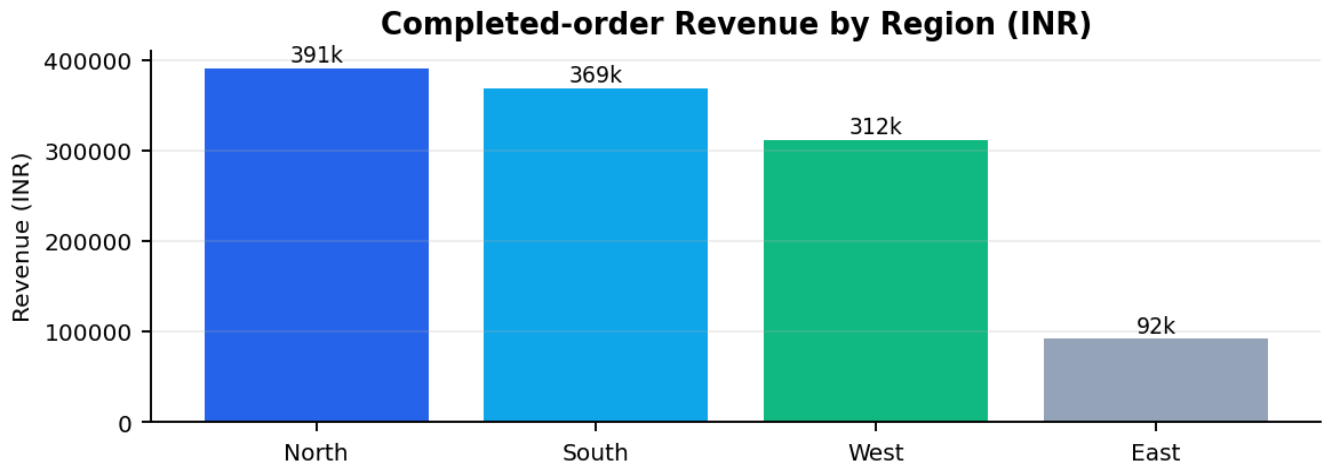
Revenue peaks in 2025-10 at INR 123,896. The lowest month is 2025-06 at INR 56,500. Because the dataset covers one simulated year, these changes demonstrate trend analysis rather than a forecast.



Category interpretation

- Electronics leads revenue at INR 372,757, but its 30.2% margin is below the overall margin.
- Lifestyle contributes the least revenue at INR 46,182, while recording the highest category margin at 45.0%.
- Revenue concentration should be considered together with profitability. High revenue does not automatically mean the highest margin.

3. Regional and Product Performance



North leads completed-order revenue at INR 391,129; East is the lowest at INR 92,401. The dataset does not include traffic, stock availability, delivery time or acquisition cost, so it cannot explain the regional difference.

Top five products by revenue

Rank	Product	Category	Units	Revenue	Profit
1	Travel Backpack	Fashion	79	INR 165,475	INR 56,455
2	USB-C Hub	Electronics	96	INR 164,264	INR 50,984
3	Bluetooth Headphones	Electronics	69	INR 154,688	INR 42,908
4	Electric Kettle	Home & Kitchen	95	INR 151,041	INR 52,241
5	Table Lamp	Home & Kitchen	96	INR 132,512	INR 44,192

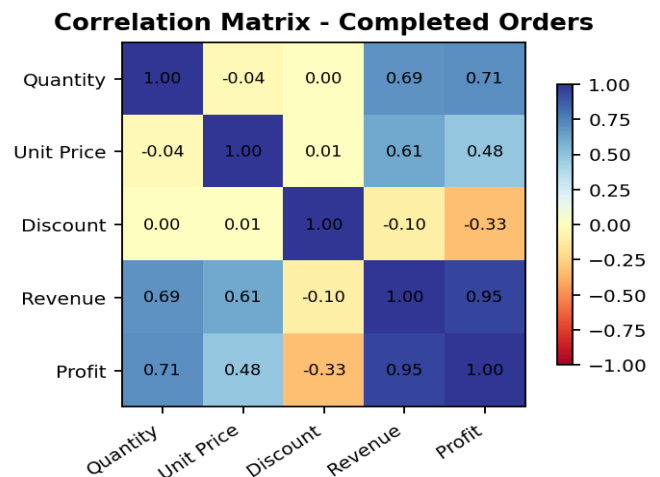
Business interpretation

- Travel Backpack generated the highest product revenue: INR 165,475.
- The strongest products can be prioritised for inventory monitoring and promotional tests when real stock and demand information becomes available.
- Lower-performing regions and categories require investigation before action. Revenue alone does not identify the cause of weaker performance.

Important: These observations show how an analyst converts numbers into business questions. They are not recommendations for Bluestock operations because the dataset is simulated.

4. Statistical Analysis and Recommendations

Metric	Completed-order revenue
Mean	INR 3,605
Median	INR 2,971
Mode	INR 2,796
Standard deviation	INR 2,535
IQR high-value threshold	INR 9,943
High-value outliers	9 orders



Statistical interpretation

- Revenue and profit move closely together in this sample, with a correlation of 0.95.
- Quantity has a 0.71 correlation with profit. Discount has a -0.33 correlation with profit.
- The IQR method identified 9 high-value orders above INR 9,943. They were retained because they are valid records, not automatic errors.

Recommended next actions

- Continue using the Completed-order rule consistently in Excel, SQL, Python and future reports.
- Investigate the reasons behind Returned and Cancelled orders, which represent 35.4% of all records in this sample.
- Monitor revenue and margin together by product and category rather than relying on revenue alone.
- Review high-value orders individually before deciding whether they are exceptional business events or data errors.

Method and limitations

The analysis uses Python with Pandas, NumPy and Matplotlib. The dataset contains 500 cleaned records covering January to December 2025. Correlation describes association, not causation. Results should be replaced with authorised company data if a dataset is later provided for this Week 1 task.